

SAKINA RIZVI

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📍 Mumbai

🌐 LinkedIn

🐙 GitHub

📁 Portfolio

Computer Engineering student with cross-disciplinary experience spanning embedded IoT hardware and full-stack AI software - taking sensor-driven prototypes from circuit to cloud by combining ESP32-based hardware nodes with Python/AI backends and modern web dashboards, alongside pure-software experience in Generative AI, NLP, and computer vision.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript

Embedded / IoT Hardware: ESP32, MPU6050, BMP280, Ultrasonic Sensors, NRF24L01 Wireless Mesh, Sensor Prototyping

AI / Machine Learning: Scikit-learn, Isolation Forest, XGBoost, Generative AI, Computer Vision, NLP, RAG Pipelines

Backend / Frontend: FastAPI, Flask, SQLite, ChromaDB, React.js, Tailwind CSS, Leaflet.js (GIS)

Data Analysis: Pandas, NumPy, EDA, Matplotlib, Seaborn, Tableau

Soft Skills: Leadership, Problem Solving, Attention to Detail, Collaboration, Communication

PROFESSIONAL EXPERIENCE

Tech Intern — Neeyat AI

Mar 2026 – Aug 2026

Diagnosed and resolved critical bottlenecks across AI tooling and workflows, enhancing system efficiency through creative troubleshooting.

Gen AI Intern — Cyart

Jul 2025 – Jan 2026

Co-engineered core components of Text-to-Speech (TTS) pipelines, supporting cross-functional team projects from development to deployment.

Data Analyst Intern — Treadalytrix

Jun 2025 – Jul 2025

Translated raw business data into strategic insights by designing data models and performing rigorous statistical analysis to guide executive decisions.

KEY PROJECTS

MineSentinel AI - (Hardware + Software)

Co-developed an AI-enabled, low-cost real-time mine subsidence early-warning system for underground coal mines

Built ESP32 multi-sensor nodes (MPU6050 tilt/vibration, BMP280, ultrasonic) with NRF24L01 mesh comms,

LED/buzzer alerts, and Li-Po power management.

Implemented a two-stage prediction pipeline (Isolation Forest + XGBoost) via a FastAPI/Flask + SQLite backend, and a

React/TypeScript/Tailwind/Leaflet.js dashboard for GIS risk-zone visualization and real-time alerts.

VisionVault — Offline-First Personal RAG Knowledge System

Built a modular, fully local RAG pipeline (PyMuPDF ingestion, chunking, BAAI/bge-small-en-v1.5 embeddings,

ChromaDB search), architected for offline operation with a planned Ollama + FastAPI/React layer.

Improved Text-to-Speech (TTS) System

Cloned and extended the VALL-E-X model, adding multi-voice detection, synthesis-error identification, and a voice/text accuracy-scoring module.

Person Detection (FDFR) & Loan Eligibility / Housing EDA

Trained/fine-tuned an RT-DETR model for real-time person detection; separately built a loan-approval classifier and ran

EDA on housing data to surface key price drivers.

Certification

Artificial Intelligence & Machine Learning
Pregrad

Generative AI
Nasscom

App Development Bootcamp
Shree L.R. Tiwari college of Engineering

Python Course
ShreeL.R. Tiwari college of Engineering

Applied AI Engineer
Surfboard Ventures / edQuest, 2026

EDUCATION

Shree L.R.TiwariCollegeof Engineering,Mumbai

2024 – Present

Bachelor of Engineering in Computer Engineering | CGPA (SE): 8.93

Royal CollegeofArts,Science andCommerce,Mumbai

2022 – 2024

HSC (Science Stream) | Percentage: 93.4%

INTERESTS

Reading · Travel · Music · Cricket · Basketball